

EE-2003 Computer Organization and Assembly Language

Saturday, February 25, 2023

Course Instructor

Mr. Ali Raza

Serial No:

1st Sessional Exam

Total Time: 1 Hour

Total Marks: 100

Signature of Invigilator

Roll No

Section

Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Verify at the start of the exam that you have a total of **four (4)** questions printed on **five (5)** pages including this title page.
2. Attempt all questions on the question-book and in the given order.
3. The exam is closed books, closed notes. Please see that the area in your threshold is free of any material classified as 'useful in the paper' or else there may a charge of cheating.
4. Read the questions carefully for clarity of context and understanding of meaning and make assumptions wherever required, for neither the invigilator will address your queries, nor the teacher/examiner will come to the examination hall for any assistance.
5. Write Anything you learned apart from COAL to get one bonus mark.
6. Fit in all your answers in the provided space. You may use extra space on the last page if required. If you do so, clearly mark question/part number on that page to avoid confusion.
7. **Use only your own stationery and calculator.** If you do not have your own calculator, use manual calculations.
8. Use only permanent ink-pens. Only the questions attempted with permanent ink-pens will be considered. Any part of paper done in lead pencil cannot be claimed for checking/rechecking.

	Q-1	Q-2	Q-3	Q-4	Total
Total Marks	10	12	18	10	50
Marks Obtained					

Vetted By: _____ **Vetter Signature:** _____

Question 1 [10 Marks]

Consider the following declaration and fill register after every instruction. Consider all registers assigned to zero in the start of the code. Memory starts at **0X0100**.

<pre>.data L3s=5 L3 dd 0abcdefh,1,2, 4,5,6 L3s db lengthof L3 L5 WORD 1,2 L6 db sizeof L3 dup(type L5 dup(1)) L7 dw L3s dup (2)</pre>	
mov AL, type L3	AL : 04
mov AH, sizeof L3	AH : 24
mov BL, lengthof L3	BL : 6
mov SI, OFFSET L3	SI : 0100
mov BH, BYTE PTR L3+1	BH : cd
mov BX, [SI+1]	BX : abcd
mov CX, L5	CX : 0001
mov BX, word ptr L6	BX : 0101
mov DL, L6[4]	DL : 01
mov BX, [2+SI]	BX : 00ab

Question 2 [7+5= 12 Marks]

Part A

Consider the following data declaration and code instructions [2+2+3]

```
.data  
bLval1 LABEL BYTE  
wLval1 LABEL WORD  
val1 DWORD 089ABCDEFH  
wLval2 LABEL WORD  
val2 BYTE 5  
val3 BYTE 6  
  
.code  
mov ax,0  
mov al,bLval1  
mov bx,wLval1  
mov cl,BYTE PTR val1  
mov dx,WORD PTR val1  
mov si,WORD PTR [val1+2]  
mov ax,wLval2
```

(a) What is the size of data segment for this program?

11 Bytes.

(b) Fill the following memory locations. Assume that your memory starts with 0x0000. Use little endian to store data.

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	inc	inc	inc	EF	CD	AB	89	inc	inc	05	06					
0010																

(c) Write instruction by using base + displacement to access value the third byte of val1 variable.

Mov al , [val1 + 2]

Part B (3 + 2 = 5 Marks)

a) What are the components of central processing unit? Briefly describe them.

- ALU
- Control Unit
- Registers
- Internal CPU interconnections

b)

i) Which of the following bus systems is used to designate the source or destination of the data on the bus itself?

a- control bus
c-address bus

b-data bus
d- system bus

ii) The main advantage of multiple bus organization over single bus is **Reduction in number of cycles** _____

- a) Reduction in the number of cycles for execution
- b) Increase in size of the registers
- c) Better Connectivity
- d) None of the mentioned

Question 3 [3*6 = 18 Marks]

Write output of the following program. If there is an error in any line of the code, mention error of the instruction and **remove error of instruction**.

Note: Assume we can have a multi-digit output as a value (not ascii).

<pre> model small .stack 100h .data Arr1 db 23,4,22,54 Arr2 DD 2,4,1234578h, 9,1 Arr3 DW AB12h, 4312h, 91, 1234h .code Mov dx,0 Mov si, Offset Arr1 Mov bx, [si] Add bx,7 ### Output ### Mov dl,bl Mov ah,02h Int 21h ### Output ### Mov si , 2 Mov Bx,0 Mov bx , Byte PTR Arr2[si*Type Arr2 + 1] ### Output ### Mov dx,bx Mov ah,02h Int 21h ### Output ### Mov esi,0 Mov bx , Word Ptr Arr2[esi] ### Output ### Mov dx,bx Mov ah,02h Int 21h ### Output ### </pre>	<pre> Mov dx,0 Mov esi, Offset Arr3 Mov bx, [esi] Add bx,7 ### Output ### Mov dl,bl Mov dx,bx Mov ah,02h Int 21h ### Output ### Mov si , Offset Arr3 Mov bx , Word PTR [si+4] // Error Mov bx,[si+4] ### Output ### Mov dx,bx Mov ah,02h Int 21h ### Output ### Mov esi,0 Mov bx , DWord Ptr Arr3[esi] // Error Mov ebx, Dword Ptr Arr3 [esi] ### Output ### Mov dx,bx /error Mov edx,ebx Mov ah,02h Int 21h ### Output ### </pre>
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Question 4 [10 Marks]

Write an assembly program which evaluates the following expression

$$C = (A - BX + B) - (D + CX) + C$$

Print C

Note: **A, B, C** and **D** are **memory variables** that should be **declared** and taken as an input from user.

```
.model small  
.stack 100h  
.data
```

```
A DB ?  
B DB ?  
C DB ?  
D DB ?
```

```
.code
```

```
Mov ax,@data  
Mov ds,ax
```

```
Mov ah,01h  
Int 21h  
Sub al,48  
Mov A,al
```

```
Mov ah,01h  
Int 21h  
Sub al,48  
Mov B,al
```

```
Mov ah,01h  
Int 21h  
Sub al,48  
Mov C,al
```

```
Mov ah,01h  
Int 21h  
Sub al,48  
Mov D,al
```

```
Mov AX,B  
Sub A,BX  
Add A,AX
```

```
Add CX, D  
Sub A,CX
```

```
Mov Ax,A  
Add C,AX
```

```
Mov DX,0  
Mov |DL,C  
Mov ah,02h  
In 21h
```